

2. Starting from a blank file, copy the code of the game from this book by manually typing it yourself. (Don't use copy and paste!)
3. Run the program again, and go back and fix any typos or bugs you may have introduced.
4. Run the program under a debugger, so you can carefully execute each line of code one at a time to understand what it does.
5. Find the comments marked with (!) to find code that you can modify and then see how this affects the program the next time you run it.
6. Finally, try to re-create the program yourself from scratch. It doesn't have to be an exact copy; you can put your own spin on the program.

When copying the code from this book, you don't necessarily have to type the comments (the text at the end of a line following the # symbol), as these are notes for human programmers and are ignored by Python. However, try to write your Python code on the same line numbers as the programs in this book to make comparison between the two easier. If you have trouble finding typos in your program, you can compare your code to the code in this book with the online diff tool at <https://inventwithpython.com/bigbookpython/diff/>.

Each program has been given a set of tags to describe it, such as **board game**, **simulation**, **artistic**, and **two-player**. An explanation of each of these tags and a cross-index of tags and projects can be found in Appendix A. The projects are listed in alphabetical order, however.

Downloading and Installing Python

Python is the name of both the programming language and the interpreter software that runs your Python code. The Python software is completely free to download and use. You can check if you already have Python installed from a command line window. On Windows, open the Command Prompt program and then enter **py --version**. If you see output like the following, then Python is installed:

```
C:\Users\Al>py --version
Python 3.9.1
```

On macOS and Linux, open the Terminal program and then enter **python3 --version**. If you see output like the following, then Python is installed:

```
$ python3 --version
Python 3.9.1
```

This book uses Python version 3. Several backward-incompatible changes were made between Python 2 and 3, and the programs in this book require at least Python version 3.1.1 (released in 2009) to run. If you see an error message telling you that Python cannot be found or the version reports Python 2, you can download the latest Python installer for your